

SHREYAS SUNIL GAIKWAD

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EDUCATION

The University of Texas at Austin

Ph.D. in Computational Science, Engineering, and Mathematics (CSEM), GPA: 4.0/4.0 2020–2025

M.S. in Computational Science, Engineering, and Mathematics (CSEM), GPA: 4.0/4.0 2019–2021

Indian Institute of Technology (IIT) Bombay, India

B.Tech with Honors in Mechanical Engineering (Minor in Computer Science), Rank: 4/150, GPA: 9.32/10.0 2015–2019

WORK EXPERIENCE

- **Lead Software R&D Engineer, Cadence Design Systems** 2026–Present
 - Contributing to **C++** platform software for a **multi-decabillion-gate** hardware emulation system (Palladium/Protium) used to validate computer chip designs before physical manufacturing
 - Diagnosing and resolving urgent customer-reported issues (**P0/P1**), including subtle edge cases in concurrent, non-deterministic logic, for high-performance, low-latency graph-traversal code underlying real-time hardware emulation
 - Developing Google Test-based unit test suite of **200+** cases covering optimization logic and multi-threaded task-bucketing
- **Graduate Research Assistant, UT Austin** 2020–2025
 - Ph.D. Thesis: Paleo-reconstruction of Greenland Ice Sheet Dynamics over the Holocene via Bayesian Inversion*
 - Developed a scalable **Bayesian inversion** framework for the Fortran-based ice sheet model SICOPOLIS using open-source **Algorithmic Differentiation (AD)** tools for efficient gradient computation [III][3]
 - Solved high-dimensional inverse problems constrained by partial differential equations (PDEs), estimating $\mathcal{O}(10^6)$ uncertain parameters in a continental-scale ice sheet model, accelerated using preconditioned quasi-Newton methods
 - Reduced sensitivity analysis cost from $\mathcal{O}(N^3)$ to $\mathcal{O}(1)$ model evaluations using adjoint-based gradients
 - Developed a novel interpretation technique for neural networks using physics-informed adjoints and **eXplainable-AI (XAI)** methods in **Keras**, validating learned insights against physically consistent adjoint sensitivities [1]
- **Machine Learning Intern, Ansys Inc.** Summer 2024
 - Built a unified **Neural Radiance Field (NeRF)** architecture to implicitly represent multiple, diverse object geometries
 - Developed a scalable **GenAI** pipeline for image-to-3D synthesis in **PyTorch** (**106 million parameters**), combining the unified NeRF architecture with a pre-trained, 3D-aware, **multi-view 2D diffusion model** for 3D asset generation
 - Trained the NeRF without 3D data using 3D-aware diffusion priors and **Score Distillation Sampling (SDS)** loss
- **Visiting Scholar, Argonne National Laboratory** Summer 2022
 - Developed the **first open-source**, **MPI-parallel**, **AD-based** Bayesian optimization framework for the atmospheric and ocean general circulation model MITgcm, an alternative to a proprietary tool costing $\sim$ **\$15,000/year per user** [II][2]

RESEARCH PROJECTS

- **Machine Learning Applications in Geophysics** Spring 2021
 - Developed CNNs & U-Nets in **Keras** for earthquake (**96% accuracy**) & seismic fault (**97% accuracy**) detection
- **Physics-Informed Machine Learning** Spring 2021
 - Trained a deep neural network in **PyTorch** to emulate and assimilate a nonlinear PDE-based mountain glacier model by leveraging higher-order derivatives from PyTorch’s computational graph
- **Laplacian 2D Finite Difference (FD) Solver Application** Fall 2020
 - Features: OOP (**C++**); solvers (Gauss, Jacobi, PETSc); build (Autotools); testing (Bats, Travis CI, Docker); **98% coverage** (lcov); **0% memory errors** (Valgrind); HPC environment (SLURM); parsing & logging (GRVY)

OPEN-SOURCE CONTRIBUTIONS

- I **diffblas**, open-source differentiated BLAS routines from their implementation in LAPACK using AD tool Tapenade
- II **MITgcm-AD v2**, open-source adjoint-based data assimilation framework for a general circulation model
- III **SICOPOLIS-AD v2**, open-source adjoint-based data assimilation framework for an ice sheet model
- IV **ARGOVIS**, developed OOP-based Python tools for interactive plotting and binning of Argo ocean data

TECHNICAL SKILLS

- **Languages:** Python (PyTorch, Keras, Tensorflow), Julia, C/C++, Fortran-77/90, MATLAB
- **HPC toolkit:** OpenMP, MPI, CUDA, SLURM, git, docker, shell scripts, CI, autotools, valgrind, lcov, GRVY, Tapenade

SELECT JOURNAL AND CONFERENCE ARTICLES

1. H. Pillar*, **S.S. Gaikwad*** et al., “Pairing eXplainable AI with Adjoint Modeling for Flexible Investigation and Robust Attribution of Ocean Variability”, *In preparation for Geophysical Research Letters* (2026).
2. **S.S. Gaikwad** et al., “MITgcm-AD v2: Open source tangent linear and adjoint modeling framework for the oceans and atmosphere enabled by the AD tool Tapenade”, *Future Generation Computer Systems* (2024).
3. **S.S. Gaikwad** et al., “SICOPOLIS-AD v2: tangent linear and adjoint modeling framework for ice sheet modeling enabled by automatic differentiation tool Tapenade”, *Journal of Open Source Software* (2023).

* Equal contributions to this manuscript.